

Risques climatiques — M2 Actuariat

Parametric uncertainty in integrated assessment models

Gauthier Vermandel

Sorbonne Université — M2 Actuariat

Wednesday, 9 September 2026, 10:45–12:45

Objectives

- Understanding the role of parametric uncertainty quantification in IAMs.

Materials:

- DICE_Notebook1-3_Uncertainty.ipynb *notebook exercises about the quantification of parametric uncertainty.*

Objectives

- Understanding the role of parametric uncertainty quantification in IAMs.

Materials:

- DICE_Notebook1-3_Uncertainty.ipynb *notebook exercises about the quantification of parametric uncertainty.*

Introduction

Introduction: Why uncertainty matters

- Climate policy is a **risk management problem** (IPCC, WG3 Ch.2 Kunreuther et al. (2014)).
- Large uncertainty in key drivers:
 - Climate sensitivity (ECS), damages, mitigation costs.
 - Socioeconomic pathways (SSPs, population, growth).
- Standard DICE baseline: *deterministic path*.
- Reality: **distribution of possible futures**

Introduction: Why uncertainty matters

- Climate policy is a **risk management problem** (IPCC, WG3 Ch.2 Kunreuther et al. (2014)).
- Large uncertainty in key drivers:
 - Climate sensitivity (ECS), damages, mitigation costs.
 - Socioeconomic pathways (SSPs, population, growth).
- Standard DICE baseline: *deterministic path*.
- Reality: **distribution of possible futures**

Introduction: Why uncertainty matters

- Climate policy is a **risk management problem** (IPCC, WG3 Ch.2 Kunreuther et al. (2014)).
- Large uncertainty in key drivers:
 - Climate sensitivity (ECS), damages, mitigation costs.
 - Socioeconomic pathways (SSPs, population, growth).
- Standard DICE baseline: *deterministic path*.
- Reality: **distribution of possible futures**

Introduction: Why uncertainty matters

- Climate policy is a **risk management problem** (IPCC, WG3 Ch.2 Kunreuther et al. (2014)).
- Large uncertainty in key drivers:
 - Climate sensitivity (ECS), damages, mitigation costs.
 - Socioeconomic pathways (SSPs, population, growth).
- Standard DICE baseline: *deterministic path*.
- Reality: **distribution of possible futures**

Introduction: Why uncertainty matters

- Climate policy is a **risk management problem** (IPCC, WG3 Ch.2 Kunreuther et al. (2014)).
- Large uncertainty in key drivers:
 - Climate sensitivity (ECS), damages, mitigation costs.
 - Socioeconomic pathways (SSPs, population, growth).
- Standard DICE baseline: *deterministic path*.
- Reality: **distribution of possible futures**

Introduction: Why uncertainty matters

- Climate policy is a **risk management problem** (IPCC, WG3 Ch.2 Kunreuther et al. (2014)).
- Large uncertainty in key drivers:
 - Climate sensitivity (ECS), damages, mitigation costs.
 - Socioeconomic pathways (SSPs, population, growth).
- Standard DICE baseline: *deterministic path*.
- Reality: **distribution of possible futures**

Introduction: Why uncertainty matters

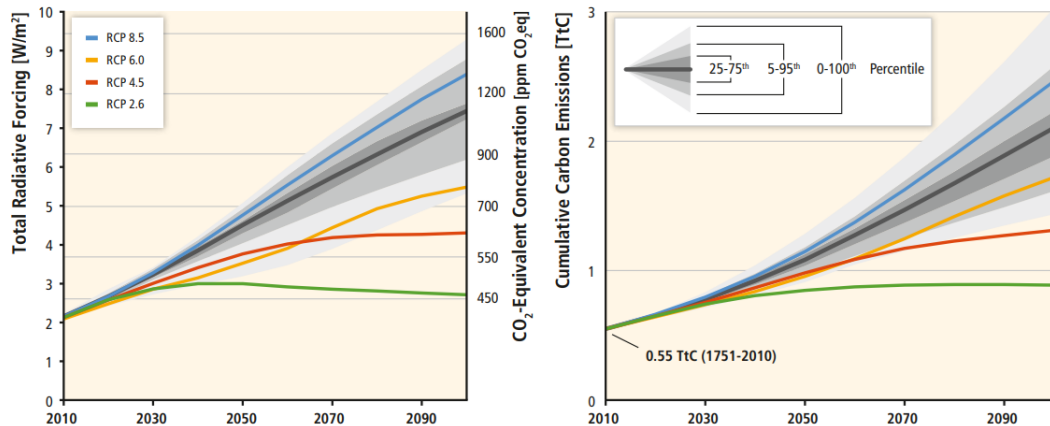


Figure 2.3 | Total radiative forcing (left panel) and cumulative carbon emissions since 1751 (right panel) in baseline scenario literature compared to RCP scenarios. Forcing was estimated ex-post from models with full coverage using the median output from the MAGICC results. Secondary axis in the left panel expresses forcing in CO₂eq concentrations. Scenarios are depicted as ranges with median emboldened; shading reflects interquartile range (darkest), 5th–95th percentile range (lighter), and full extremes (lightest). Source: Figure 6.6 from WGIII AR5.

Introduction: Taxonomy of uncertainty

- **Parametric**: uncertain parameter values (ECS, damages).
- **Paradigmatic (or model-structural) uncertainty**: different IAM structures, climate models.
- **Scenario**: socioeconomic, policy, emissions pathways.
- **Trend vs. cycle**: long-run drivers vs short-term shocks.
- **Ambiguity**: situations where probabilities themselves are unclear or contested.

Introduction: Taxonomy of uncertainty

- **Parametric**: uncertain parameter values (ECS, damages).
- **Paradigmatic (or model-structural) uncertainty**: different IAM structures, climate models.
- **Scenario**: socioeconomic, policy, emissions pathways.
- **Trend vs. cycle**: long-run drivers vs short-term shocks.
- **Ambiguity**: situations where probabilities themselves are unclear or contested.

Introduction: Taxonomy of uncertainty

- **Parametric**: uncertain parameter values (ECS, damages).
- **Paradigmatic (or model-structural) uncertainty**: different IAM structures, climate models.
- **Scenario**: socioeconomic, policy, emissions pathways.
- **Trend vs. cycle**: long-run drivers vs short-term shocks.
- **Ambiguity**: situations where probabilities themselves are unclear or contested.

Introduction: Taxonomy of uncertainty

- **Parametric**: uncertain parameter values (ECS, damages).
- **Paradigmatic (or model-structural) uncertainty**: different IAM structures, climate models.
- **Scenario**: socioeconomic, policy, emissions pathways.
- **Trend vs. cycle**: long-run drivers vs short-term shocks.
- **Ambiguity**: situations where probabilities themselves are unclear or contested.

Introduction: Taxonomy of uncertainty

- **Parametric**: uncertain parameter values (ECS, damages).
- **Paradigmatic (or model-structural) uncertainty**: different IAM structures, climate models.
- **Scenario**: socioeconomic, policy, emissions pathways.
- **Trend vs. cycle**: long-run drivers vs short-term shocks.
- **Ambiguity**: situations where probabilities themselves are unclear or contested.

IPCC guidance: two complementary scales

- **Confidence** = level of certainty in a finding
 - Combines: **evidence** (amount, quality, consistency of data) $\times$ **agreement** (consensus across studies).
 - Expressed qualitatively: low / medium / high / very high confidence.
 - Example: "High confidence that warming exceeds natural variability."
- **Likelihood** = assessed probability of an outcome
 - Mapped to calibrated terms:
 - Very likely = $> 90\%$, likely = $> 66\%$, about as likely as not = $33\text{--}66\%$, etc.
 - Example: "It is *likely* that ECS is between $1.5\text{--}4.5^{\circ}\text{C}$."

In modeling practice:

- Report probability distributions, fan charts, quantile bands.
- Highlight **tail risks**: low-probability but high-impact outcomes.

IPCC guidance: two complementary scales

- **Confidence** = level of certainty in a finding
 - Combines: **evidence** (amount, quality, consistency of data) $\times$ **agreement** (consensus across studies).
 - Expressed qualitatively: low / medium / high / very high confidence.
 - Example: "High confidence that warming exceeds natural variability."
- **Likelihood** = assessed probability of an outcome
 - Mapped to calibrated terms:
 - Very likely = $> 90\%$, likely = $> 66\%$, about as likely as not = $33\text{--}66\%$, etc.
 - Example: "It is *likely* that ECS is between $1.5\text{--}4.5^\circ\text{C}$."

In modeling practice:

- Report probability distributions, fan charts, quantile bands.
- Highlight **tail risks**: low-probability but high-impact outcomes.

IPCC guidance: two complementary scales

- **Confidence** = level of certainty in a finding
 - Combines: **evidence** (amount, quality, consistency of data) $\times$ **agreement** (consensus across studies).
 - Expressed qualitatively: low / medium / high / very high confidence.
 - Example: "High confidence that warming exceeds natural variability."
- **Likelihood** = assessed probability of an outcome
 - Mapped to calibrated terms:
 - Very likely = $> 90\%$, likely = $> 66\%$, about as likely as not = $33\text{--}66\%$, etc.
 - Example: "It is *likely* that ECS is between $1.5\text{--}4.5^\circ\text{C}$."

In modeling practice:

- Report probability distributions, fan charts, quantile bands.
- Highlight **tail risks**: low-probability but high-impact outcomes.

IPCC guidance: two complementary scales

- **Confidence** = level of certainty in a finding
 - Combines: **evidence** (amount, quality, consistency of data) $\times$ **agreement** (consensus across studies).
 - Expressed qualitatively: low / medium / high / very high confidence.
 - Example: "High confidence that warming exceeds natural variability."
- **Likelihood** = assessed probability of an outcome
 - Mapped to calibrated terms:
 - Very likely = $> 90\%$, likely = $> 66\%$, about as likely as not = $33\text{--}66\%$, etc.
 - Example: "It is *likely* that ECS is between $1.5\text{--}4.5^\circ\text{C}$."

In modeling practice:

- Report probability distributions, fan charts, quantile bands.
- Highlight **tail risks**: low-probability but high-impact outcomes.

IPCC guidance: two complementary scales

- **Confidence** = level of certainty in a finding
 - Combines: **evidence** (amount, quality, consistency of data) $\times$ **agreement** (consensus across studies).
 - Expressed qualitatively: low / medium / high / very high confidence.
 - Example: "High confidence that warming exceeds natural variability."
- **Likelihood** = assessed probability of an outcome
 - Mapped to calibrated terms:
 - Very likely = $> 90\%$, likely = $> 66\%$, about as likely as not = 33–66%, etc.
 - Example: "It is *likely* that ECS is between 1.5–4.5°C."

In modeling practice:

- Report probability distributions, fan charts, quantile bands.
- Highlight **tail risks**: low-probability but high-impact outcomes.

IPCC guidance: two complementary scales

- **Confidence** = level of certainty in a finding
 - Combines: **evidence** (amount, quality, consistency of data) $\times$ **agreement** (consensus across studies).
 - Expressed qualitatively: low / medium / high / very high confidence.
 - Example: "High confidence that warming exceeds natural variability."
- **Likelihood** = assessed probability of an outcome
 - Mapped to calibrated terms:
 - Very likely = $> 90\%$, likely = $> 66\%$, about as likely as not = $33\text{--}66\%$, etc.
 - Example: "It is *likely* that ECS is between $1.5\text{--}4.5^\circ\text{C}$."

In modeling practice:

- Report probability distributions, fan charts, quantile bands.
- Highlight **tail risks**: low-probability but high-impact outcomes.

IPCC guidance: two complementary scales

- **Confidence** = level of certainty in a finding
 - Combines: **evidence** (amount, quality, consistency of data) $\times$ **agreement** (consensus across studies).
 - Expressed qualitatively: low / medium / high / very high confidence.
 - Example: "High confidence that warming exceeds natural variability."
- **Likelihood** = assessed probability of an outcome
 - Mapped to calibrated terms:
 - Very likely = $> 90\%$, likely = $> 66\%$, about as likely as not = 33–66%, etc.
 - Example: "It is *likely* that ECS is between 1.5–4.5°C."

In modeling practice:

- Report probability distributions, fan charts, quantile bands.
- Highlight **tail risks**: low-probability but high-impact outcomes.

IPCC guidance: two complementary scales

- **Confidence** = level of certainty in a finding
 - Combines: **evidence** (amount, quality, consistency of data) $\times$ **agreement** (consensus across studies).
 - Expressed qualitatively: low / medium / high / very high confidence.
 - Example: "High confidence that warming exceeds natural variability."
- **Likelihood** = assessed probability of an outcome
 - Mapped to calibrated terms:
 - Very likely = $> 90\%$, likely = $> 66\%$, about as likely as not = 33–66%, etc.
 - Example: "It is *likely* that ECS is between 1.5–4.5°C."

In modeling practice:

- Report probability distributions, fan charts, quantile bands.
- Highlight **tail risks**: low-probability but high-impact outcomes.

IPCC guidance: two complementary scales

- **Confidence** = level of certainty in a finding
 - Combines: **evidence** (amount, quality, consistency of data) $\times$ **agreement** (consensus across studies).
 - Expressed qualitatively: low / medium / high / very high confidence.
 - Example: "High confidence that warming exceeds natural variability."
- **Likelihood** = assessed probability of an outcome
 - Mapped to calibrated terms:
 - Very likely = $> 90\%$, likely = $> 66\%$, about as likely as not = 33–66%, etc.
 - Example: "It is *likely* that ECS is between 1.5–4.5°C."

In modeling practice:

- Report probability distributions, fan charts, quantile bands.
- Highlight **tail risks**: low-probability but high-impact outcomes.

IPCC guidance: two complementary scales

- **Confidence** = level of certainty in a finding
 - Combines: **evidence** (amount, quality, consistency of data) $\times$ **agreement** (consensus across studies).
 - Expressed qualitatively: low / medium / high / very high confidence.
 - Example: "High confidence that warming exceeds natural variability."
- **Likelihood** = assessed probability of an outcome
 - Mapped to calibrated terms:
 - Very likely = $> 90\%$, likely = $> 66\%$, about as likely as not = 33–66%, etc.
 - Example: "It is *likely* that ECS is between 1.5–4.5°C."

In modeling practice:

- Report probability distributions, fan charts, quantile bands.
- Highlight **tail risks**: low-probability but high-impact outcomes.

Introduction: Focus in this course: parametric uncertainty

- Treat parameters as **random variables**.
- Examples in DICE from Nordhaus (2018):
 - Equilibrium climate sensitivity (T_{2xCO_2}).
 - Damage function coefficients (a_2, a_3).
 - Backstop technology cost (p_b).
- Goal: generate an ensemble of plausible trajectories.

Introduction: Focus in this course: parametric uncertainty

- Treat parameters as **random variables**.
- Examples in DICE from **Nordhaus (2018)**:
 - Equilibrium climate sensitivity (T_{2xCO_2}).
 - Damage function coefficients (a_2, a_3).
 - Backstop technology cost (p_b).
- Goal: generate an ensemble of plausible trajectories.

Introduction: Focus in this course: parametric uncertainty

- Treat parameters as **random variables**.
- Examples in DICE from Nordhaus (2018):
 - Equilibrium climate sensitivity (T_{2xCO_2}).
 - Damage function coefficients (a_2, a_3).
 - Backstop technology cost (p_b).
- Goal: generate an ensemble of plausible trajectories.

Introduction: Focus in this course: parametric uncertainty

- Treat parameters as **random variables**.
- Examples in DICE from Nordhaus (2018):
 - Equilibrium climate sensitivity (T_{2xCO_2}).
 - Damage function coefficients (a_2, a_3).
 - Backstop technology cost (p_b).
- Goal: generate an ensemble of plausible trajectories.

Introduction: Focus in this course: parametric uncertainty

- Treat parameters as **random variables**.
- Examples in DICE from **Nordhaus (2018)**:
 - Equilibrium climate sensitivity (T_{2xCO_2}).
 - Damage function coefficients (a_2, a_3).
 - Backstop technology cost (p_b).
- Goal: generate an ensemble of plausible trajectories.

Introduction: Focus in this course: parametric uncertainty

- Treat parameters as **random variables**.
- Examples in DICE from **Nordhaus (2018)**:
 - Equilibrium climate sensitivity (T_{2xCO_2}).
 - Damage function coefficients (a_2, a_3).
 - Backstop technology cost (p_b).
- Goal: generate an ensemble of plausible trajectories.

Parametric Uncertainty

Parametric Uncertainty: Sampling the population of calibrations

Monte Carlo sampling

- Draw N parameter sets from a prior (here: truncated Gaussian for a_2).
- Each draw $\Rightarrow$ one implied outcome; in a full run, one DICE trajectory.
- Contrast *priors* (literature-based) vs *posteriors* (after calibration/estimation).

Very simple illustration (single-parameter): quadratic damages

$$\text{Damage factor at temperature } T : \frac{1}{1 + a_2 T^2}$$

- Sample $a_2 \sim \mathcal{N}^+(0.00236, 0.001^2)$ (truncated at $a_2 > 0$), fix $T = 3^\circ\text{C}$.
- Report empirical quantiles (5–95%) for a_2 and for the damage factor.
- Visuals: histogram of a_2 , histogram of damage factor, and a_2 vs damage-factor scatter.

Parametric Uncertainty: Sampling the population of calibrations

Monte Carlo sampling

- Draw N parameter sets from a prior (here: truncated Gaussian for a_2).
- Each draw $\Rightarrow$ one implied outcome; in a full run, one DICE trajectory.
- Contrast *priors* (literature-based) vs *posteriors* (after calibration/estimation).

Very simple illustration (single-parameter): quadratic damages

$$\text{Damage factor at temperature } T : \frac{1}{1 + a_2 T^2}$$

- Sample $a_2 \sim \mathcal{N}^+(0.00236, 0.001^2)$ (truncated at $a_2 > 0$), fix $T = 3^\circ\text{C}$.
- Report empirical quantiles (5–95%) for a_2 and for the damage factor.
- Visuals: histogram of a_2 , histogram of damage factor, and a_2 vs damage-factor scatter.

Parametric Uncertainty: Sampling the population of calibrations

Monte Carlo sampling

- Draw N parameter sets from a prior (here: truncated Gaussian for a_2).
- Each draw $\Rightarrow$ one implied outcome; in a full run, one DICE trajectory.
- Contrast *priors* (literature-based) vs *posteriors* (after calibration/estimation).

Very simple illustration (single-parameter): quadratic damages

$$\text{Damage factor at temperature } T : \frac{1}{1 + a_2 T^2}$$

- Sample $a_2 \sim \mathcal{N}^+(0.00236, 0.001^2)$ (truncated at $a_2 > 0$), fix $T = 3^\circ\text{C}$.
- Report empirical quantiles (5–95%) for a_2 and for the damage factor.
- Visuals: histogram of a_2 , histogram of damage factor, and a_2 vs damage-factor scatter.

Parametric Uncertainty: Sampling the population of calibrations

Monte Carlo sampling

- Draw N parameter sets from a prior (here: truncated Gaussian for a_2).
- Each draw $\Rightarrow$ one implied outcome; in a full run, one DICE trajectory.
- Contrast *priors* (literature-based) vs *posteriors* (after calibration/estimation).

Very simple illustration (single-parameter): quadratic damages

$$\text{Damage factor at temperature } T : \frac{1}{1 + a_2 T^2}$$

- Sample $a_2 \sim \mathcal{N}^+(0.00236, 0.001^2)$ (truncated at $a_2 > 0$), fix $T = 3^\circ\text{C}$.
- Report empirical quantiles (5–95%) for a_2 and for the damage factor.
- Visuals: histogram of a_2 , histogram of damage factor, and a_2 vs damage-factor scatter.

Parametric Uncertainty: Sampling the population of calibrations

Monte Carlo sampling

- Draw N parameter sets from a prior (here: truncated Gaussian for a_2).
- Each draw $\Rightarrow$ one implied outcome; in a full run, one DICE trajectory.
- Contrast *priors* (literature-based) vs *posteriors* (after calibration/estimation).

Very simple illustration (single-parameter): quadratic damages

$$\text{Damage factor at temperature } T : \frac{1}{1 + a_2 T^2}$$

- Sample $a_2 \sim \mathcal{N}^+(0.00236, 0.001^2)$ (truncated at $a_2 > 0$), fix $T = 3^\circ\text{C}$.
- Report empirical quantiles (5–95%) for a_2 and for the damage factor.
- Visuals: histogram of a_2 , histogram of damage factor, and a_2 vs damage-factor scatter.

Parametric Uncertainty: Sampling the population of calibrations

Monte Carlo sampling

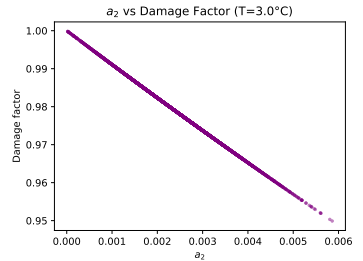
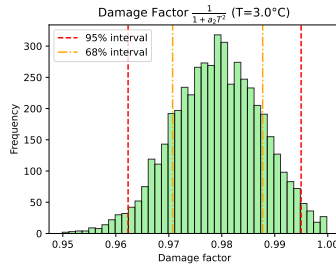
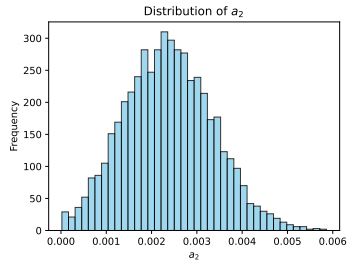
- Draw N parameter sets from a prior (here: truncated Gaussian for a_2).
- Each draw $\Rightarrow$ one implied outcome; in a full run, one DICE trajectory.
- Contrast *priors* (literature-based) vs *posteriors* (after calibration/estimation).

Very simple illustration (single-parameter): quadratic damages

$$\text{Damage factor at temperature } T : \frac{1}{1 + a_2 T^2}$$

- Sample $a_2 \sim \mathcal{N}^+(0.00236, 0.001^2)$ (truncated at $a_2 > 0$), fix $T = 3^\circ\text{C}$.
- Report empirical quantiles (5–95%) for a_2 and for the damage factor.
- Visuals: histogram of a_2 , histogram of damage factor, and a_2 vs damage-factor scatter.

Parametric Uncertainty: Illustration



- Distributions summarize **parametric uncertainty**.
- Quantile bands and fan charts generalize to full DICE trajectories (temperature, welfare, carbon price).
- Not a point prediction: a **range of plausible outcomes**; policy should pay attention to tails.

- Distributions summarize **parametric uncertainty**.
- Quantile bands and fan charts generalize to full DICE trajectories (temperature, welfare, carbon price).
- Not a point prediction: a **range of plausible outcomes**; policy should pay attention to tails.

- Distributions summarize **parametric uncertainty**.
- Quantile bands and fan charts generalize to full DICE trajectories (temperature, welfare, carbon price).
- Not a point prediction: a **range of plausible outcomes**; policy should pay attention to tails.

Example: damage function

- At $T = 3^{\circ}\text{C}$, the quadratic damage factor $\frac{1}{1+a_2T^2}$ varies substantially across draws.
- In our Monte Carlo exercise:
 - 68% interval $\approx [0.971, 0.988]$
 - 95% interval $\approx [0.962, 0.995]$
- Interpretation: depending on a_2 , GDP losses at 3°C range from $\sim 3.8\%$ to $\sim 0.5\%$. $\Rightarrow$ **Uncertainty quantification of damage parameter translates into uncertainty on damages themselves.**

Example: damage function

- At $T = 3^{\circ}\text{C}$, the quadratic damage factor $\frac{1}{1+a_2T^2}$ varies substantially across draws.
- In our Monte Carlo exercise:
 - 68% interval $\approx [0.971, 0.988]$
 - 95% interval $\approx [0.962, 0.995]$
- Interpretation: depending on a_2 , GDP losses at 3°C range from $\sim 3.8\%$ to $\sim 0.5\%$. $\Rightarrow$ **Uncertainty quantification of damage parameter translates into uncertainty on damages themselves.**

Example: damage function

- At $T = 3^{\circ}\text{C}$, the quadratic damage factor $\frac{1}{1+a_2T^2}$ varies substantially across draws.
- In our Monte Carlo exercise:
 - 68% interval $\approx [0.971, 0.988]$
 - 95% interval $\approx [0.962, 0.995]$
- Interpretation: depending on a_2 , GDP losses at 3°C range from $\sim 3.8\%$ to $\sim 0.5\%$. $\Rightarrow$ **Uncertainty quantification of damage parameter translates into uncertainty on damages themselves.**

Example: damage function

- At $T = 3^{\circ}\text{C}$, the quadratic damage factor $\frac{1}{1+a_2T^2}$ varies substantially across draws.
- In our Monte Carlo exercise:
 - 68% interval $\approx [0.971, 0.988]$
 - 95% interval $\approx [0.962, 0.995]$
- Interpretation: depending on a_2 , GDP losses at 3°C range from $\sim 3.8\%$ to $\sim 0.5\%$. $\Rightarrow$ **Uncertainty quantification of damage parameter translates into uncertainty on damages themselves.**

Example: damage function

- At $T = 3^{\circ}\text{C}$, the quadratic damage factor $\frac{1}{1+a_2T^2}$ varies substantially across draws.
- In our Monte Carlo exercise:
 - 68% interval $\approx [0.971, 0.988]$
 - 95% interval $\approx [0.962, 0.995]$
- Interpretation: depending on a_2 , GDP losses at 3°C range from $\sim 3.8\%$ to $\sim 0.5\%$. $\Rightarrow$ **Uncertainty quantification of damage parameter translates into uncertainty on damages themselves.**

Two levels of uncertainty:

- **Uncertainty quantification (UQ):**
 - Given model structure, explore parametric distributions (Monte Carlo, ensembles, fan charts).
 - Output: ranges of outcomes (temperature, damages, welfare).
- **Uncertainty in the planner's problem:**
 - Planner internalizes future risks $\Rightarrow$ **precautionary motive**.
 - Early work: Cai and Lontzek (2019), stochastic DICE.
 - Uncertainty directly affects optimal abatement, carbon tax.

Two levels of uncertainty:

- **Uncertainty quantification (UQ):**
 - Given model structure, explore parametric distributions (Monte Carlo, ensembles, fan charts).
 - Output: ranges of outcomes (temperature, damages, welfare).
- **Uncertainty in the planner's problem:**
 - Planner internalizes future risks $\Rightarrow$ precautionary motive.
 - Early work: Cai and Lontzek (2019), stochastic DICE.
 - Uncertainty directly affects optimal abatement, carbon tax.

Two levels of uncertainty:

- **Uncertainty quantification (UQ):**
 - Given model structure, explore parametric distributions (Monte Carlo, ensembles, fan charts).
 - Output: ranges of outcomes (temperature, damages, welfare).
- **Uncertainty in the planner's problem:**
 - Planner internalizes future risks $\Rightarrow$ precautionary motive.
 - Early work: Cai and Lontzek (2019), stochastic DICE.
 - Uncertainty directly affects optimal abatement, carbon tax.

Two levels of uncertainty:

- **Uncertainty quantification (UQ):**
 - Given model structure, explore parametric distributions (Monte Carlo, ensembles, fan charts).
 - Output: ranges of outcomes (temperature, damages, welfare).
- **Uncertainty in the planner's problem:**
 - Planner internalizes future risks $\Rightarrow$ **precautionary motive**.
 - Early work: **Cai and Lontzek (2019)**, stochastic DICE.
 - Uncertainty directly affects optimal abatement, carbon tax.

Two levels of uncertainty:

- **Uncertainty quantification (UQ):**
 - Given model structure, explore parametric distributions (Monte Carlo, ensembles, fan charts).
 - Output: ranges of outcomes (temperature, damages, welfare).
- **Uncertainty in the planner's problem:**
 - Planner internalizes future risks $\Rightarrow$ **precautionary motive**.
 - Early work: Cai and Lontzek (2019), stochastic DICE.
 - Uncertainty directly affects optimal abatement, carbon tax.

Two levels of uncertainty:

- **Uncertainty quantification (UQ):**
 - Given model structure, explore parametric distributions (Monte Carlo, ensembles, fan charts).
 - Output: ranges of outcomes (temperature, damages, welfare).
- **Uncertainty in the planner's problem:**
 - Planner internalizes future risks $\Rightarrow$ **precautionary motive**.
 - Early work: **Cai and Lontzek (2019)**, stochastic DICE.
 - Uncertainty directly affects optimal abatement, carbon tax.

Two levels of uncertainty:

- **Uncertainty quantification (UQ):**
 - Given model structure, explore parametric distributions (Monte Carlo, ensembles, fan charts).
 - Output: ranges of outcomes (temperature, damages, welfare).
- **Uncertainty in the planner's problem:**
 - Planner internalizes future risks $\Rightarrow$ **precautionary motive**.
 - Early work: **Cai and Lontzek (2019)**, stochastic DICE.
 - Uncertainty directly affects optimal abatement, carbon tax.

Challenges:

- Stochastic DICE $\Rightarrow$ high-dimensional state space (climate, capital, shocks).
- Requires advanced computational methods (stochastic dynamic programming, adaptive sparse grids, GPUs).
- Trade-off: richer treatment of uncertainty vs. feasibility for teaching / policy use.

Challenges:

- Stochastic DICE $\Rightarrow$ high-dimensional state space (climate, capital, shocks).
- Requires advanced computational methods (stochastic dynamic programming, adaptive sparse grids, GPUs).
- Trade-off: richer treatment of uncertainty vs. feasibility for teaching / policy use.

Challenges:

- Stochastic DICE $\Rightarrow$ high-dimensional state space (climate, capital, shocks).
- Requires advanced computational methods (stochastic dynamic programming, adaptive sparse grids, GPUs).
- Trade-off: richer treatment of uncertainty vs. feasibility for teaching / policy use.

Thank you!

`gauthier.vermandel@polytechnique.edu`

References

- Cai, Y. and Lontzek, T. (2019). The social cost of carbon with economic and climate risks. *Journal of Political Economy*, 127:2684–2734.
- Kunreuther, H., Gupta, S., Bosetti, V., Cooke, R., Dutt, V., Ha-Duong, M., Held, H., Llanes-Regueiro, J., Patt, A., Shittu, E., et al. (2014). Integrated risk and uncertainty assessment of climate change response policies. In *Climate Change 2014: Mitigation of Climate Change: Working Group III Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*, pages 151–206. Cambridge University Press.
- Nordhaus, W. (2018). Projections and uncertainties about climate change in an era of minimal climate policies. *American Economic Journal: Economic Policy*, 10:333–360.